MORA

NOCA

Following climate-predicted direction

Changing the climate-community lag

← against following →

← narrowing widening →

Pinus albicaulis

Abies lasiocarpa

Picea engelmannii

Tsuga mertensiana

Pinus monticola

Pseudotsuga menziesii

Callitropsis nootkatensis

Abies amabilis

Tsuga heterophylla

Pinus contorta

Abies procera

Abies grandis

Thuja plicata

Pinus ponderosa

-0.10 -0.05 0.00 0.05 0.10

-0.25 0.00 0.25

Species contribution

Maximum elevation (m)

1,300 1,500 1,700 1,900 2,100

Holocene stability periods

Holocene climate-change periods

Anthropocene